



HEILONGJIANG WANGYUNFENG WIND POWER PROJECT

Document Prepared by Longyuan (Beijing) Carbon Asset Management
Technology Co., Ltd.

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1 PROJECT DETAILS

1.1 Summary Description of the Implementation Status of the Project

Heilongjiang Wangyunfeng Wind Power Project (hereinafter refers to the project) is developed by Hegang Longyuan Wind Power Co., Ltd. and located on Damahe Forest Farm, Luobei County, Hegang City, Heilongjiang Province, P.R.China. Totally 58 sets of 850kW wind turbines were installed, providing a total capacity of 49.3MW. It is estimated the power supply was 120,069 MWh annually and achieved amount of 123,430 tCO₂e emission reduction every year as per registered PD. The objective of the project is to generate electricity using state-of-the-art wind power generation technology and to sell into Northeast China Power Grid (NECPG).

Construction start permission of the project was issued on 04-Jun-2009 and the construction contract was signed on 05-Jun-2009. The first wind turbine started to commercial operation on 20-Jan-2010 and all wind turbines started commercial operation on 12-Apr-2010.

This is the 2nd VCS monitoring period from 01-Jan-2013 to 31-Dec-2018 under VCS program. During this monitoring period, the operation condition of all the wind turbines and the operation environment haven't been changed from the previous monitoring periods, which are also consistent with the description of the registered PD, the project has generated the total net electricity supply of 503,067.772 MWh and achieved 517,149 tCO₂e emission reduction.

1.2 Sectoral Scope and Project Type

Sectoral scope: 1 energy industries (Renewable sources).

Project type: Grid-connected wind power project.

The project is not a grouped project..

1.3 Project Proponent

Organization name	Hegang Longyuan Wind Power Co., Ltd.
Contact person	Mr. Zhang Yulei
Title	Marketing Manager
Address	17th Committee, Fengxiang Town, Luobei County, Hegang City, Heilongjiang Province, China
Telephone	+8645151919150
Email	yulei.zhang.c@chnenergy.com.cn

1.4 Other Entities Involved in the Project

Organization name	Longyuan (Beijing) Carbon Asset Management Technology Co., Ltd.
Role in the Project	Consultancy
Contact person	Shanshan Yu
Title	General Manager
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Email	20069189@ceic.com

1.5 Project Start Date

20-Jan-2010 (the first wind turbine started to commercial operation).

1.6 Project Crediting Period

The first 10-year crediting period of the project under VCS is from 20-Jan-2010 to 19-Jan-2020 (both days included).

1.7 Project Location

The project is located in Damahe Forest Farm, Luobei County, Hegang City, Heilongjiang Province of China. Its central geographical coordinates are north latitude 47°54'34", and east longitude 130°39'56", its altitude is about 240m~430m. The detailed location of the project is shown in the Figure 1-1 below.



Figure 1-1 Location of the project

1.8 Title and Reference of Methodology

Title: Approved consolidated baseline and monitoring methodology ACM0002 “Consolidated methodology for grid-connected electricity generation from renewable sources”- ACM0002 (Version 12.1.0)

The methodology also refers to the latest approved versions of the following tools:

- ◆ Tool to calculate the emission factor for an electricity system (Version 02.2.1);
- ◆ Tool for the demonstration and assessment of additionality (Version 05.2.1).

<https://cdm.unfccc.int/methodologies/PAmethodologies/approved>

1.9 Participation under other GHG Programs

The project was registered as a CDM project on 31-Jan-2012 under the UNFCCC reference number 5711 with a first 7-year CDM renewable crediting period from 31-Jan-2012 to 30-Jan-2019, the detailed information can be found at <https://cdm.unfccc.int/Projects/DB/BVQI1327230197.63/view>

Start Date	End Date	GHG Emission Reduction (tCO ₂ e)	Credit	Program
20-Jan-2010	30-Jan-2012	195,043	VCU	VCS
31-Jan-2012	31-Dec-2012	72,155	CER	CDM

This is the second monitoring period under VCS program from 01-Jan-2013 to 31-Dec-2018 for the project. All emission reductions during this monitoring period have not and will not seek CDM CER issuance. Emission reductions during this monitoring period will only seek issuance under VCS program.

1.10 Other Forms of Credit

Emission Trading Programs and Other Binding Limits

China has a national emissions trading scheme only cover the high-emission industries, such as thermal power generation, petrochemical, chemical, building materials, iron and steel, non-ferrous, paper, aviation and other key emission industries that emitted at least 26,000 tons of CO₂e/year, not including renewable project¹.

¹ http://www.mee.gov.cn/xxgk2018/xxgk/xxgk05/202103/t20210330_826728.html

The project proponent Hegang Longyuan Wind Power Co., Ltd., as an enterprise for renewable energy investment, is not included in the compliance entity list by China national Emission Trading Scheme (ETS). Also, the project has not been registered as a CCER (Chinese Certified Emission Reductions) project in China, thus it is not eligible for emission reductions transaction under the China's ETS. Therefore, the project does not reduce GHG emissions from activities that are included in an emissions trading program or any other mechanism that includes GHG allowance trading. The net GHG emission reductions generated during this monitoring period have not been used for compliance under such programs or mechanisms.

Furthermore, as a voluntary emission reduction project, the GHG emission reductions and removals generated by the project will not be enforced to be used for compliance in China. A statement on no double counting will be submitted to Verra to confirm the credits during this monitoring period has not been counted and will not be counted under emission trading programs and other binding limits.

Other Forms of Environmental Credit

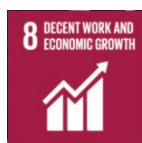
The project has not sought or received another form of GHG-related environmental credit, including renewable energy certificates, during this monitoring period.

1.11 Sustainable Development Contributions

The project activity is a renewable wind power project located in Heilongjiang Province, Northeast China.



The project utilizes wind resources to generate and supply 503,067.772 MWh renewable electricity to NECPG, which contributes to SDG 7.



The project provides 12 long-term job opportunities for local, which has a positive effect on the local economy and contributes to SDG 8.



The project reduces 517,149 tCO₂e GHG emission and contributes to SDG 13.

These contribute to achieve China's stated sustainable development priorities, included strengthening resistance capacity to climate risks, increasing labor force participation rate, increasing the share of clean energy consumption.

Table 1: Sustainable Development Contributions

Row number	SDG Target	SDG Indicator	Net Impact on SDG Indicator	Current Project Contributions	Contributions Over Project Lifetime
1)	7.2	Renewable energy supplied to the power grid	Implemented activities to increase	503,067.772 MWh of electricity from renewable sources has been supplied to the power grid during the reporting period	An accumulated 692,799.292 MWh of electricity from renewable sources has been supplied to the power grid by end of the reporting period
2)	8.5	Numbers of job opportunities provided by implementation of the project	Implemented activities to increase	12 people have been employed by the project as long-term employee whose salary higher above the national average line level	12 people have been employed by the project as long-term employee whose salary higher above the national average line level by end of the reporting period
3)	13.0	Tonnes of greenhouse gas emissions avoided	Implemented activities to increase	The project has achieved a total GHG emission reduction of 517,149 tCO ₂ e during the reporting period.	An accumulated 712,192 tCO ₂ e of GHG emission reduction have been achieved by the project by end of the reporting period

2 SAFEGUARDS

2.1 No Net Harm

Environmental Impact

The Environmental Impact Assessment (EIA) of the project has been conducted and approved by local government officials. The EIA report has assessed every possible aspect of environmental impacts of the project and recommended corresponding measures, where applicable. The environmental impacts and corresponding mitigation measures during operation have been discussed in the registered PD. No negative environmental impacts have been identified.

Socio-economic impact

Furthermore, the project is renewable project and achieves the SDG 7 and SDG 13. Also, the project provides 12 long-term job opportunities for local, which has a positive effect on the local economy and contributes to SDG 8. Please refer to section 1.11 of this report for details.

In conclusion, no net harm on local environment and social community has been detected for the project.

2.2 Local Stakeholder Consultation

LSC prior to the project implementation

Local stakeholder consultation has been conducted on 06-May-2009. All comments received have been well addressed prior to the project implementation. Please refer to section E of the registered PD (version 3 dated 21-Dec-2012) of the project for detailed information.

In conclusion, there was no significant negative comments received in the project preparation stage.

LSC during the operation period

Communications with Local stakeholders have been carried out at periodic intervals. Key implementation schedules and changes of the project have been communicated to the local authority, the neighbourhood committee, and local residents. The comments and suggestions from residents were collected by the local authority, if any.

The project owner carried out questionnaire survey for the local stakeholders to collect the relevant comments and suggestions every two years, e.g., in May 2014, July 2016 and May 2018 during this monitoring period. The local authority also conducts spot checks on the implementation of the project at periodic intervals as per relevant regulations. There have been no negative comments received for the project so far.

In line with VCS requirements all the processes have been implemented to receive comments from local stakeholders as well as communicate with them at periodic intervals.

2.3 AFOLU-Specific Safeguards

The project is a non-AFOLU project, this section is not required.

3 IMPLEMENTATION STATUS

3.1 Implementation Status of the Project Activity

Totally 58 sets of wind turbines with a total capacity of 49.3MW were installed and supply an average annual generation of 120,069MWh to NECPG. All wind turbines are produced in China. Technical Characteristics of Wind Turbines is shown as below table 3-1.

Each wind turbine has a box transformer which transforms the voltage to 35kV from which the 35kV lines link into the on-site 110kV switchgear at the project site. By the 110kV line, the electricity generated by the project is delivered to the power grid. The wind turbines and transmission facility could be monitored and controlled by onsite central control room. The technologies employed in the project activity are advanced domestic technologies, which is no technology transfer activity involved. The main advantage of this domestic manufactured turbine is its' adaptability to different types of wind resources and its improved generation efficiency, so it has been widely installed in China.

A technology diagram of the project is presented as Figure 4-2 in section 4.3.

During this monitoring period, the project was implemented in accordance with the registered PD. The project has operated without any accidental or emergency events that might impact the GHG emission reductions or removals and monitoring.

Main technical information of wind turbines used in the project

Table 3-1 Technical Characteristics of Wind Turbines²

Parameters	Data	
Rated wind speed (m/s)	13	
Unit capacity	850kW	
Type	G58-850	G52-850
Diameter of rotor (m)	58	52

² Sourced from Wind Turbines Purchase Agreement

Cut-in wind speed (m/s)	3
Cut-out wind speed (m/s)	21
Survival wind speed (m/s)	37.5
Swept area (m ²)	2642/2214
Lifetime	20 year
Generator	Doubly fed asynchronous generator

3.2 Deviations

3.2.1 Methodology Deviations

There is no methodology deviation in this monitoring period.

3.2.2 Project Description Deviations

There is a deviation for the crediting period. The project is registered under VCS 3.2 and completed validation before 19-Mar-2020, thus it remains eligible to apply the crediting period requirements under VCS Version 3 which shall be a maximum of ten years and may be renewed at most twice, so the first renewable crediting period of the project shall be updated from 20-Jan-2010~30-Jan-2012 to 20-Jan-2010~19-Jan-2020. This monitoring period belongs to the first crediting period from 20-Jan-2010~19-Jan-2020 (10 years). The project did not complete the validation of the crediting period update within two years of the end of the first crediting period, therefore the VCS total crediting period ends on 19-Jan-2020, the end of the first crediting period.

The deviation does not impact the applicability of the methodology, additionality and the appropriateness of the baseline scenario.

3.3 Grouped Projects

Not applicable.

4 DATA AND PARAMETERS

4.1 Data and Parameters Available at Validation

Data / Parameter	EF _y
Data unit	tCO ₂ /MWh
Description	Combined margin CO ₂ emission factor for grid connected power generation in the year y
Source of data	Registered PD

Value applied	1.0280
Justification of choice of data or description of measurement methods and procedures applied	Ex-ante determined in the registered PD and fixed during the 1 st crediting period.
Purpose of Data	Calculation of baseline emissions
Comments	Determined ex-ante and fixed for the 1 st crediting period

4.2 Data and Parameters Monitored

Data / Parameter	EG _{facility, y}																	
Data unit	MWh/yr																	
Description	Quantity of net electricity generation supplied by the project plant/unit to the grid in year y																	
Source of data	Measured by electricity meters: M1																	
Description of measurement methods and procedures to be applied	The main meter M1 with accuracy of 0.5S installed at outlet of the main transformer and the meter M2 with accuracy of 0.5S installed at the inlet of the main transformer measure the quantity of the electricity exported to the grid by the project and the quantity of electricity imported from the grid by the project. Quantity of net electricity generation supplied by the project to the grid is the difference between the quantity of the electricity exported to the grid by the project and the quantity of electricity imported from the grid by the project.																	
Frequency of monitoring/recording	The reading of electricity meters are continuously measured, daily read and monthly aggregated.																	
Value monitored	<table><tr><td>Period</td><td>Value</td></tr><tr><td>01-Jan-2013~31-Dec-2013</td><td>94,884.240</td></tr><tr><td>01-Jan-2014~31-Dec-2014</td><td>88,372.944</td></tr><tr><td>01-Jan-2015~31-Dec-2015</td><td>62,437.620</td></tr><tr><td>01-Jan-2016~31-Dec-2016</td><td>82,676.880</td></tr><tr><td>01-Jan-2017~31-Dec-2017</td><td>85,289.848</td></tr><tr><td>01-Jan-2018~31-Dec-2018</td><td>89,406.240</td></tr><tr><td>Total</td><td>503,067.772</td></tr></table>		Period	Value	01-Jan-2013~31-Dec-2013	94,884.240	01-Jan-2014~31-Dec-2014	88,372.944	01-Jan-2015~31-Dec-2015	62,437.620	01-Jan-2016~31-Dec-2016	82,676.880	01-Jan-2017~31-Dec-2017	85,289.848	01-Jan-2018~31-Dec-2018	89,406.240	Total	503,067.772
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QA/QC procedures to be applied	The meters (accuracy degree is not less than 0.5%, bidirectional) are calibrated yearly according to national standards and rules. Double checked by receipt of sales. The designated person of the project reads and aggregates the number of the meter. Data is archived for 2 years following the end of the																	

	crediting period by means of electronic and paper backup.
Purpose of the data	Calculation of baseline emissions
Calculation method	$EG_{\text{facility}, y} = EG_{\text{export}, y} - EG_{\text{import}, y}$
Comments	-

Data / Parameter	$EG_{\text{export}, y}$																
Data unit	MWh																
Description	Electricity exported to the grid by the project																
Source of data	Measured by electricity meters: M1																
Description of measurement methods and procedures to be applied	The main meter M1 with accuracy of 0.5S installed at outlet of the main transformer and the meter M2 with accuracy of 0.5S installed at the inlet of the main transformer measure the quantity of the electricity exported to the grid by the project and the quantity of electricity imported from the grid by the project.																
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Monitoring equipment	See as follow table 4-1																
QA/QC procedures to be applied	The meters (accuracy degree is not less than 0.5%, bidirectional) are calibrated yearly according to national standards and rules. Double checked by receipt of sales. The designated person of the project reads and aggregates the number of the meter. Data is archived for 2 years following the end of the crediting period by means of electronic and paper backup.																
Purpose of the data	Calculation of baseline emissions																
Calculation method	-																
Comments	-																

Data / Parameter	EG _{import, y}																
Data unit	MWh																
Description	Electricity utilized by the project																
Source of data	Measured by electricity meters: M1																
Description of measurement methods and procedures to be applied	The main meter M1 with accuracy of 0.5S installed at outlet of the main transformer and the meter M2 with accuracy of 0.5S installed at the inlet of the main transformer measure the quantity of the electricity exported to the grid by the project and the quantity of electricity imported from the grid by the project.																
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Purpose of the data	Calculation of baseline emissions																
Calculation method	-																
Comments	-																

4.3 Monitoring Plan

Introduction

The project adopts the approved consolidated baseline and monitoring methodology ACM0002 to determine the emission reductions from the wind farm. This plan describes in more detail the process.

Responsibility

Hegang Longyuan Wind Power Co., Ltd. is overall responsible for monitoring and carries out the monitoring following this monitoring plan. The VCS manager of Hegang Longyuan Wind Power Co., Ltd. is responsible for the monitoring and reporting of the wind farm. The management structure is illustrated as follows:

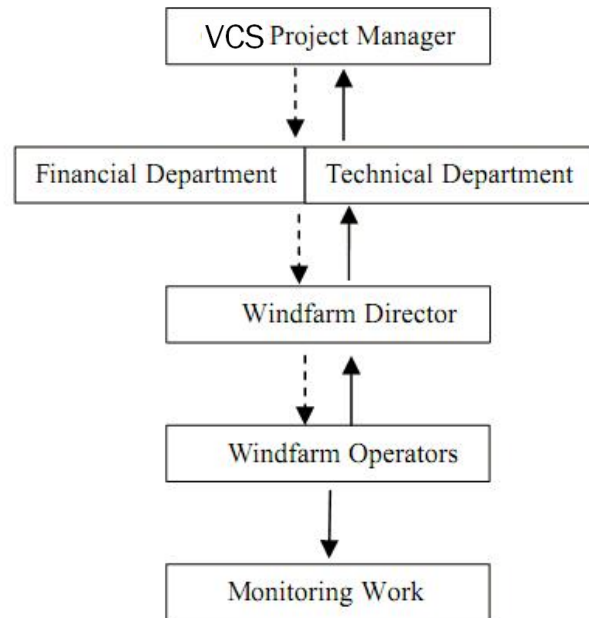


Figure 4-1 Organizational structure

Windfarm Director collects the information and data required by the Monitoring Plan. The collected information is documented and sent to the VCS manager and responsible staffs of Hegang Longyuan Wind Power Co., Ltd. monthly. The VCS manager is in charge of the implementation of the Monitoring Plan and the confirmations on monitoring, calculation data and report to the General Manager of the company. Managers of the project must maintain credible, transparent, and adequate data estimation, measurement, collection, and tracking systems to maintain the information required for an audit of an emission reduction project.

Monitoring equipment

Two bi-directional electricity meters are installed at the on-site 110 kV substation (the main meter M1 with accuracy of 0.5S installed at outlet of the main transformer and the meter M2 with accuracy of 0.5S installed at the inlet of the main transformer) to monitor directly $EG_{import,y}$ and $EG_{export,y}$ by the project (shown in the figure below). During the monitoring period, the main meter (M1) and the backup meter (M2) all worked normally. Therefore, the monitored data used in the MR were all sourced from the main meter (M1).

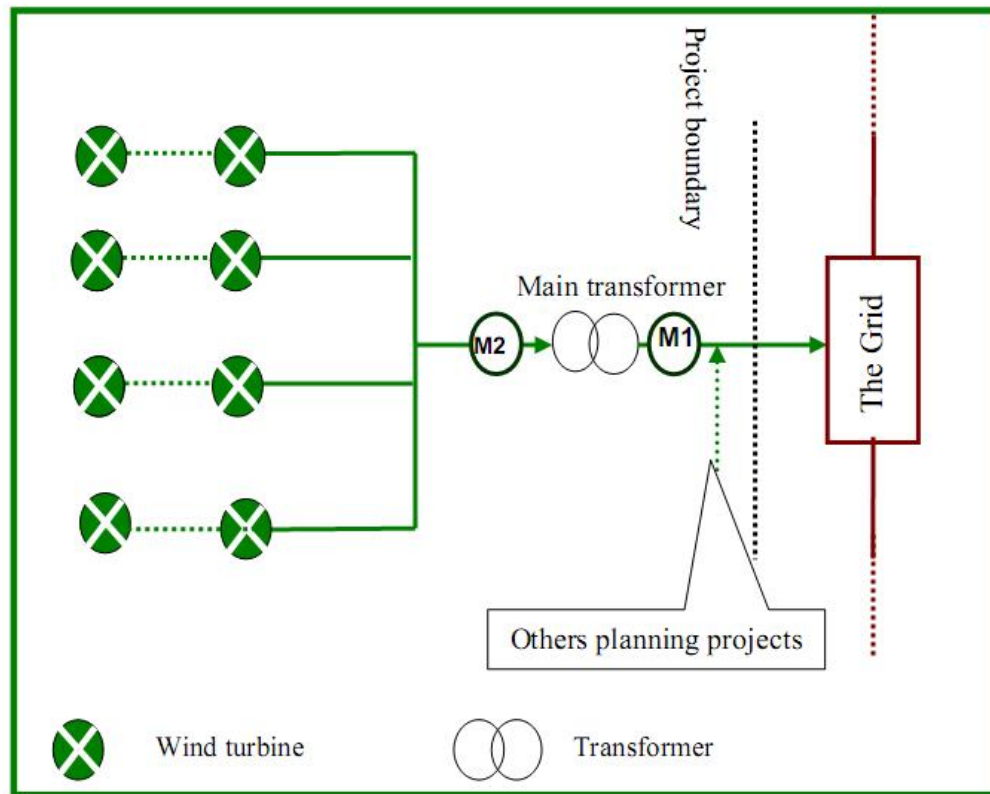


Figure 4-2 A technology diagram of the project

Calibration

The metering equipments related with the project are calibrated and tested yearly by a qualified third party appointed by the Northeast China Power Grid for accuracy according to the requirement from Technical administrative code of electric energy metering. The meters shall be jointly inspected and sealed on behalf of the parties concerned and shall not be interfered with by either party except in the presence of the other party or its accredited representatives.

Table 4-1 Monitoring Equipments of the Project

Meter	Type	Serial number	Calibrate Date	Validity date	Accuracy	Calibration Frequency	Calibration Entity
Main meter M1	DTSD719	4200525040909208867	12-May-2012 10-May-2013 08-May-2014 06-May-2015 04-May-2016 02-May-2017 30-Apr-2018	11-May-2013 09-May-2014 07-May-2015 05-May-2016 03-May-2017 01-May-2018 29-Apr-2019	0.5S	Annually	Hegang Metrological Verification Institution
Backup meter M2	DTSD719	4200222040909208865	12-May-2012 10-May-2013 08-May-2014 06-May-2015 04-May-2016 02-May-2017 30-Apr-2018	11-May-2013 09-May-2014 07-May-2015 05-May-2016 03-May-2017 01-May-2018 29-Apr-2019	0.5S	Annually	Hegang Metrological Verification Institution

Monitoring data

According to the illustration figure above, the net electricity supplied to the grid by the project is calculated by:

$$EG_{\text{facility},y} = EG_{\text{export},y} - EG_{\text{import},y}$$

Data collection

The readings of all electricity meters are continuously measured, daily read and monthly aggregated. The designated staff collects the measured electricity data and complete the corresponding records on a monthly basis. The cut-off time of meter is at 24:00 of last day of the month.

Before being archived, these records are checked by other staffs to ensure the correctness. The data from these records is digested and analysed and the results are reported to company administrator or supervisor. Data will be archived for 2 years following the end of the last crediting period by means of electronic and paper backup.

Quality assurance and Quality control

The quality assurance and quality control procedures for the emergency about recording, maintaining and archiving data shall be improved as part of this VCS project activity. This is an on-going process that can be ensured through the VCS in terms of the need for verification of the emissions on an annual basis according to methodology ACM0002.

Hegang Longyuan Wind Power Co., Ltd specially issued the regulations of monitoring and management for VCS project (hereafter refer as VCS Manual) to monitor and verify the emission reductions from the project. Moreover, the management procedures and regulations like operation, meter recordings, maintenance and emergency management are established in the VCS Manual, which guides the technicians to operate and guarantee the quality assurance and quality control procedures for recording, maintaining and archiving data in detail. This is an on-going process that can be ensured through the VCS in terms of the need for verification of the emissions on an annual basis according to methodology ACM0002.

Several processes required to meet the requirements for emissions reduction monitoring include:

The project owner reads the meters system monthly and supplies the recordings to the local Electric Power Company. The local Electric Power Company checks the recordings supplied by the project owner.

If either party finds any reading of the Meters more than the allowable error or the monitoring equipment functions improperly, it should inform the other party immediately. The project owner and the local Power Grid Company should retain a qualified measure institute together to check the meters or equipment, solve the problems and get everything into normal condition;

In the mean time, both sides shall jointly prepare an acceptable estimation method according to Regulations and Rules of Power Supply¹ issued by State Electricity Regulatory Commission of China to calculate the electricity exported to the Grid, and provide evidence to VVB for the verification to show the estimation is reasonable and conservative.

For the handling of disputes between the project owner and the Grid, measures are adopted according to relevant articles of Interim Measures for Settlement of Electricity Charges between the Power Generating Enterprises and the Grid Enterprises issued by State Electricity Regulatory Commission of China. Moreover, if meters are out of order, the electricity generation in the period is not considered.

Data Management System

Overall responsibility for monitoring greenhouse gas emissions reductions rests with the VCS monitoring staff of the project. The VCS manual sets out the procedures for tracking information from the primary source to data calculations in paper format. Moreover, the credibility and reliability of those data and information must be confirmed. Physical documentation such as paper-based maps, diagrams and environmental assessment are collated in a central place, together with this monitoring plan. In order to facilitate auditor's reference, monitoring results are indexed. All paper-based information will be stored by Hegang Longyuan Wind Power Co., Ltd and kept at least one copy, and all data including calibration records is kept until 2 years after the end of the total crediting period of the VCS project.

The following table below outlines the key documents relevant to monitoring and verification of the emission reductions from the project.

Table 4-2 Monitoring Equipments of the Project

I.D.No.	Document Title	Main Content	Source
F-1	PDD, including the electronic spreadsheets and supporting documentation(assumptions, estimations, measurement, etc)	Calculation procedure of emission reduction and monitoring items	Project owner or UNFCC website
F-2	Report on monitoring and checking of electricity supplied to the grid	Record based on monthly meter reading and electricity sale receipts	project owner
F-3	Report on maintenance and calibration of metering equipment	Reasons for maintenance and calibration and the precision after maintenance and calibration	project owner
F-4	Report on the qualifications of the operators	Technical post ,working experience etc.	project owner
F-5	the project management record (including date collection and management system)	Comprehensively and truly reflect the management and the operation of the project	project owner

5 QUANTIFICATION OF GHG EMISSION REDUCTIONS AND REMOVALS

5.1 Baseline Emissions

According to ACM0002 and the registered PD of the Project, The baseline emission BE_y during the monitoring period results from:

$$(1) \quad BE_y = EG_{facility, y} \times EF_y = (EG_{export, y} - EG_{import, y}) \times EF_y$$

Where,

BE_y Baseline emissions of the Project in year y (t CO₂e)

EF_y Combined margin CO₂ emission factor for grid connected power generation (tCO₂e/MWh)

$EG_{facility, y}$ Net electricity supplied to the grid by the Project (MWh)

$EG_{export, y}$ Quantity of electricity exported to the grid by the Project (MWh)

$EG_{import, y}$ Quantity of electricity imported from the grid by the Project (MWh)

The monitored $EG_{facility, y}$ is reported in the table below.

Table 5-1 Monitored monthly value of $EG_{\text{facility}, y}$

	$EG_{\text{export}, y}$			$EG_{\text{import}, y}$			$EG_{\text{facility}, y}$ (MWh)
Period	Monthly Recording from main meter (M1) (MWh)	Sales Receipt (MWh)	Conservative Quantity of the electricity exported to the grid (MWh)	Monthly Recording from main meter (M1) (MWh)	Purchase Receipt (MWh)	Conservative Quantity of the electricity imported from the grid (MWh)	
	A	B	C=min(A,B)	D	E	F=max(D,E)	G=C-F
01/01/2013-31/01/2013	11,354.640	11,354.640	11,354.640	17.688	17.688	17.688	11,336.952
01/02/2013-28/02/2013	9,191.160	9,191.160	9,191.160	66.000	66.000	66.000	9,125.160
01/03/2013-31/03/2013	10,307.880	10,307.880	10,307.880	34.452	34.452	34.452	10,273.428
01/04/2013-30/04/2013	7,316.760	7,316.760	7,316.760	14.652	14.652	14.652	7,302.108
01/05/2013-31/05/2013	5,825.160	5,825.160	5,825.160	18.084	18.084	18.084	5,807.076
01/06/2013-30/06/2013	4,877.400	4,877.400	4,877.400	31.020	31.020	31.020	4,846.380
01/07/2013-31/07/2013	6,316.200	6,316.200	6,316.200	28.116	28.116	28.116	6,288.084
01/08/2013-31/08/2013	3,033.360	3,033.360	3,033.360	26.004	26.004	26.004	3,007.356
01/09/2013-30/09/2013	7,710.120	7,710.120	7,710.120	40.788	40.788	40.788	7,669.332
01/10/2013-31/10/2013	10,178.520	10,178.520	10,178.520	11.220	11.220	11.220	10,167.300
01/11/2013-30/11/2013	9,617.520	9,617.520	9,617.520	31.020	31.020	31.020	9,586.500
01/12/2013-31/12/2013	9,514.560	9,514.560	9,514.560	39.996	39.996	39.996	9,474.564
Subtotal	95,243.280	95,243.280	95,243.280	359.040	359.040	359.040	94,884.240
01/01/2014-31/01/2014	9,554.160	9,554.160	9,554.160	26.928	26.928	26.928	9,527.232
01/02/2014-28/02/2014	9,410.280	9,410.280	9,410.280	26.400	26.400	26.400	9,383.880
01/03/2014-31/03/2014	7,227.000	7,227.000	7,227.000	38.808	38.808	38.808	7,188.192
01/04/2014-30/04/2014	6,974.880	6,974.880	6,974.880	18.216	18.216	18.216	6,956.664
01/05/2014-31/05/2014	8,364.840	8,364.840	8,364.840	22.176	22.176	22.176	8,342.664

01/06/2014-30/06/2014	3,692.040	3,692.040	3,692.040	18.480	18.480	18.480	3,673.560
01/07/2014-31/07/2014	5,095.200	5,095.200	5,095.200	46.728	46.728	46.728	5,048.472
01/08/2014-31/08/2014	3,923.040	3,923.040	3,923.040	22.176	22.176	22.176	3,900.864
01/09/2014-30/09/2014	6,085.200	6,085.200	6,085.200	25.344	25.344	25.344	6,059.856
01/10/2014-31/10/2014	9,768.000	9,768.000	9,768.000	14.520	14.520	14.520	9,753.480
01/11/2014-30/11/2014	9,580.560	9,580.560	9,580.560	10.560	10.560	10.560	9,570.000
01/12/2014-31/12/2014	9,005.040	9,005.040	9,005.040	36.960	36.960	36.960	8,968.080
Subtotal	88,680.240	88,680.240	88,680.240	307.296	307.296	307.296	88,372.944
01/01/2015-31/01/2015	11,201.520	11,201.520	11,201.520	10.560	10.560	10.560	11,190.960
01/02/2015-28/02/2015	4,052.400	4,052.400	4,052.400	88.440	88.440	88.440	3,963.960
01/03/2015-31/03/2015	3,711.810	3,711.810	3,711.810	80.520	80.520	80.520	3,631.290
01/04/2015-30/04/2015	3,247.200	3,247.200	3,247.200	26.400	26.400	26.400	3,220.800
01/05/2015-31/05/2015	7,459.320	7,459.320	7,459.320	10.560	10.560	10.560	7,448.760
01/06/2015-30/06/2015	4,585.680	4,585.680	4,585.680	26.400	26.400	26.400	4,559.280
01/07/2015-31/07/2015	2,873.610	2,873.610	2,873.610	34.320	34.320	34.320	2,839.290
01/08/2015-31/08/2015	3,574.560	3,574.560	3,574.560	44.880	44.880	44.880	3,529.680
01/09/2015-30/09/2015	5,062.200	5,062.200	5,062.200	30.360	30.360	30.360	5,031.840
01/10/2015-31/10/2015	4,477.440	4,477.440	4,477.440	14.520	14.520	14.520	4,462.920
01/11/2015-30/11/2015	5,446.680	5,446.680	5,446.680	0.000	0.000	0.000	5,446.680
01/12/2015-31/12/2015	7,171.560	7,171.560	7,171.560	59.400	59.400	59.400	7,112.160
Subtotal	62,863.980	62,863.980	62,863.980	426.360	426.360	426.360	62,437.620
01/01/2016-31/01/2016	7,962.240	7,962.240	7,962.240	42.240	42.240	42.240	7,920.000
01/02/2016-29/02/2016	3,157.440	3,157.440	3,157.440	19.800	19.800	19.800	3,137.640
01/03/2016-31/03/2016	5,260.200	5,260.200	5,260.200	46.200	46.200	46.200	5,214.000
01/04/2016-30/04/2016	6,214.560	6,214.560	6,214.560	19.800	19.800	19.800	6,194.760
01/05/2016-31/05/2016	10,750.080	10,750.080	10,750.080	10.560	10.560	10.560	10,739.520
01/06/2016-30/06/2016	4,366.560	4,366.560	4,366.560	25.080	25.080	25.080	4,341.480
01/07/2016-31/07/2016	4,345.440	4,345.440	4,345.440	34.320	34.320	34.320	4,311.120
01/08/2016-31/08/2016	5,530.800	5,530.800	5,530.800	31.680	31.680	31.680	5,499.120

01/09/2016-30/09/2016	6,733.320	6,733.320	6,733.320	29.040	29.040	29.040	6,704.280
01/10/2016-31/10/2016	12,475.320	12,475.320	12,475.320	5.676	5.676	5.676	12,469.644
01/11/2016-30/11/2016	8,271.120	8,271.120	8,271.120	21.384	21.384	21.384	8,249.736
01/12/2016-31/12/2016	7,906.800	7,906.800	7,906.800	11.220	11.220	11.220	7,895.580
Subtotal	82,973.880	82,973.880	82,973.880	297.000	297.000	297.000	82,676.880
01/01/2017-31/01/2017	7,398.600	7,398.600	7,398.600	25.344	25.344	25.344	7,373.256
01/02/2017-28/02/2017	4,608.120	4,608.120	4,608.120	14.388	14.388	14.388	4,593.732
01/03/2017-31/03/2017	4,264.920	4,264.920	4,264.920	20.856	20.856	20.856	4,244.064
01/04/2017-30/04/2017	9,212.280	9,212.280	9,212.280	15.708	15.708	15.708	9,196.572
01/05/2017-31/05/2017	8,466.480	8,466.480	8,466.480	7.496	7.496	7.496	8,458.984
01/06/2017-30/06/2017	4,509.120	4,509.120	4,509.120	29.568	29.568	29.568	4,479.552
01/07/2017-31/07/2017	5,058.240	5,058.240	5,058.240	27.456	27.456	27.456	5,030.784
01/08/2017-31/08/2017	5,732.760	5,732.760	5,732.760	40.788	40.788	40.788	5,691.972
01/09/2017-30/09/2017	6,623.760	6,623.760	6,623.760	26.004	26.004	26.004	6,597.756
01/10/2017-31/10/2017	11,201.520	11,201.520	11,201.520	16.632	16.632	16.632	11,184.888
01/11/2017-30/11/2017	12,112.320	12,112.320	12,112.320	13.992	13.992	13.992	12,098.328
01/12/2017-31/12/2017	6,358.440	6,358.440	6,358.440	18.480	18.480	18.480	6,339.960
Subtotal	85,546.560	85,546.560	85,546.560	256.712	256.712	256.712	85,289.848
01/01/2018-31/01/2018	11,052.360	11,052.360	11,052.360	48.840	48.840	48.840	11,003.520
01/02/2018-28/02/2018	6,864.000	6,864.000	6,864.000	40.920	40.920	40.920	6,823.080
01/03/2018-31/03/2018	10,858.320	10,858.320	10,858.320	23.760	23.760	23.760	10,834.560
01/04/2018-30/04/2018	7,357.680	7,357.680	7,357.680	6.600	6.600	6.600	7,351.080
01/05/2018-31/05/2018	5,030.520	5,030.520	5,030.520	21.120	21.120	21.120	5,009.400
01/06/2018-30/06/2018	5,433.120	5,433.120	5,433.120	19.800	19.800	19.800	5,413.320
01/07/2018-31/07/2018	4,534.200	4,534.200	4,534.200	19.800	19.800	19.800	4,514.400
01/08/2018-31/08/2018	3,288.120	3,288.120	3,288.120	38.280	38.280	38.280	3,249.840
01/09/2018-30/09/2018	7,117.440	7,117.440	7,117.440	22.440	22.440	22.440	7,095.000
01/10/2018-31/10/2018	7,897.560	7,897.560	7,897.560	19.800	19.800	19.800	7,877.760
01/11/2018-30/11/2018	8,235.480	8,235.480	8,235.480	11.880	11.880	11.880	8,223.600

01/12/2018-31/12/2018	12,029.160	12,029.160	12,029.160	18.480	18.480	18.480	12,010.680
Subtotal	89,697.960	89,697.960	89,697.960	291.720	291.720	291.720	89,406.240
Total	505,005.900	505,005.900	505,005.900	1,938.128	1,938.128	1,938.128	503,067.772

The annual baseline emissions of the project are calculated as follows:

Table 5-2 Calculation of Baseline emissions

Year	EG _{facility, y} (MWh)	EF _y (tCO ₂ /MWh)	BE _y (tCO ₂)
	A	B	C=A*B
01-Jan-2013~31-Dec-2013	94,884.240	1.0280	97,540
01-Jan-2014~31-Dec-2014	88,372.944	1.0280	90,847
01-Jan-2015~31-Dec-2015	62,437.620	1.0280	64,185
01-Jan-2016~31-Dec-2016	82,676.880	1.0280	84,991
01-Jan-2017~31-Dec-2017	85,289.848	1.0280	87,677
01-Jan-2018~31-Dec-2018	89,406.240	1.0280	91,909
Total	503,067.772		517,149

In summary, the baseline emissions BE_y during this monitoring period is 517,149 tCO₂e.

5.2 Project Emissions

As per the applied methodology, the project emission is zero.

5.3 Leakage

As per the applied methodology, leakage of the project is not considered.

5.4 Net GHG Emission Reductions and Removals

This monitoring period started from 01-Jan-2013 to 31-Dec-2018, with totally 2,191 days. Based on the annual estimated emission reductions from the registered PD, the amount of emission reductions for this monitoring period would be $123,430 \text{ tCO}_2\text{e}/365 \text{ d} \times 2,191 \text{ d} = 740,918 \text{ tCO}_2\text{e}$. The actual emission reductions in this monitoring period (2,191 days) are 517,149 tCO₂e, which are 30.20% lower than the estimation in the registered PD. It is because wind fluctuations and the limitation of the grid absorption capacity. So, it is reasonable that during this monitoring period, the actual emission reduction is less than the estimated value.

Year	Baseline emissions or removals (tCO ₂ e)	Project emissions or removals (tCO ₂ e)	Leakage emissions (tCO ₂ e)	Net GHG emission reductions or removals (tCO ₂ e)
01-Jan-2013~31-Dec-2013	97,540	0	0	97,540
01-Jan-2014~31-Dec-2014	90,847	0	0	90,847
01-Jan-2015~31-Dec-2015	64,185	0	0	64,185
01-Jan-2016~31-Dec-2016	84,991	0	0	84,991
01-Jan-2017~31-Dec-2017	87,677	0	0	87,677
01-Jan-2018~31-Dec-2018	91,909	0	0	91,909
Total	517,149	0	0	517,149

APPENDIX1: <SUPPORTING EVIDENCE>

Evidence for SDG 7 and SDG 13: From project operation date onwards, the net electricity supplied to the grid and corresponding emission reduction in every monitoring period have been listed in below table. Please refer to following links for details.

<https://registry.verra.org/app/projectDetail/VCS/1051>

Period		EG _{facility,y}	ER	Program
From	To	MWh	tCO ₂ e	
20-Jan-2010	30-Jan-2012	189,731.520	195,043	VCS
01-Jan-2013	31-Dec-2018	503,067.772	517,149	
Total		692,799.292	712,192	

All impacts during the previous VCS monitoring period from 20-Jan-2010 to 30-Jan-2012 has been calculated in the cumulative impact. Thus, the cumulative impact has been calculated by summing the current project contributions with all impacts included in previously approved VCS monitoring reports or Sustainable Development Contribution Reports.

Evidence for SDG 8: The average annual salary of Hegang Longyuan Wind Power Co., Ltd.'s employees is shown in the material below:

鹤岗龙源风力发电有限公司员工年均工资薪金水平说明

本公司负责黑龙江望云峰风电项目的运营，2013 年-2020 年度年均工资薪金情况如下表：

年份	员工人数	年均工资（元）
2013	12	131279
2014	12	123161
2015	12	114839
2016	12	160265
2017	12	184354
2018	12	200988
2019	12	192704
2020	12	243355

特此说明。

鹤岗龙源风力发电有限公司

